
Rainbow Sunglasses Store Sales Management System

Authors

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Executive Summary

The Rainbow Sunglasses Management System developed over several project phases into a more complete and better organized database-backed web application. As our team moved from the early planning stages into implementation, we made a number of changes to improve the database design, strengthen consistency across the project materials, and make the user interface easier to use. Feedback from earlier project steps helped shape those revisions and pushed the project toward a more polished final result.

One of the biggest areas of improvement was the database schema. Earlier versions included attributes such as `salesMonthly` and `totalAmount`, but these were later removed because they were derived values and did not fit well within a cleaner relational design. We also revised the `InvoiceSunglasses` table by adding its own primary key so that update and delete operations would be easier to support in the application. In addition, we made the documentation more consistent by aligning names, capitalization, and relationship details across the overview, database outline, ERD, and schema. Foreign key behavior was also refined with `ON DELETE` handling to better represent how related records should behave.

The application itself also improved as the project progressed. Earlier versions were more limited in presentation and usability, while the final version provides a clearer interface for browsing and managing project data. Based on prior feedback, we improved readability by using clearer column aliases, displayed missing customer email values as "N/A," and reorganized the `InvoiceSunglasses` page to make the many-to-many relationship easier to understand. Update functionality was also improved through forms that auto-populate values from the selected record, which made editing more intuitive and easier to demonstrate.

Major revisions influenced by feedback included:

- removing derived attributes to improve schema design
- adding a primary key to the InvoiceSunglasses table
- standardizing naming and formatting across project materials
- improving readability through clearer column aliases
- displaying blank customer emails as “N/A”
- improving update forms with auto-populated values
- refining the layout of the many-to-many relationship page
- maintaining a working deployed version on the ENGR server

AI tools were used in a limited support role for brainstorming and occasional wording help. Final implementation and design decisions were based on our own work and the course requirements.

Overall, the final version of the Rainbow Sunglasses Management System is more consistent, more usable, and more complete than the earlier drafts.

Project Outline

Rainbow Sunglasses is a family owned retail store with 5 employees and approximately 1400 transactions a year (\$200,000 revenue). This old fashioned business uses handwritten notebooks to track sales and around 40 distinct inventory items. This manual system is slow to look through. Our team offers a modern day solution to more accurately record invoices of sunglasses to customers from employees.

This custom built website and database requires only a one time payment, while other companies require yearly subscriptions and charge fees for every transaction. Our Sunglasses Store Sales Management System allows receipts to be digitalized, the ability to quickly look up customers, and track employee performance using data insights. Owners will also be able to more accurately track inventory, thus simplifying administrative efficiency.

Database Outline

Customers

- Purpose:
 - Records core information about customers who purchase sunglasses from the store.
- Attributes:
 - customerId: int, auto_increment, not NULL, PK.
 - firstName: varchar(100), not NULL.
 - lastName: varchar(100), not NULL.
 - email: varchar(100), NULL.
 - phoneNumber: varchar(20), NOT NULL.
- Relationship:
 - 1:M between Customers and Invoices.
- Additional Details:
 - The Customers table links a customer to their Invoices of which they may have zero to many. A customer record may exist before any purchases are made.

Employees:

- Purpose:
 - Records information about staff members who create invoices and make sales.

- Attributes:
 - employeeID: int, auto_increment, not NULL, PK.
 - firstName: varchar(100), not NULL.
 - lastName: varchar(100), not NULL.
 - hireDate: date, not NULL.
 - isActive: tinyint (boolean), not NULL, default 1.
- Relationship:
 - 1:M between Employees and Invoices.
- Additional Details:
 - The Employees table links an employee to their Invoices of which they may have zero to many. An employee record may exist before any invoices are created. Digitizing employees eliminates ambiguous signatures on invoices.
 - All employees have the attribute isActive set to 1 by default, meaning they are currently employed by the management. Upon a soft deletion, the attribute will be set to 0 instead of permanently removing the employee.

Invoices

- Purpose:
 - Records each sale transaction as a single invoice.
- Attributes:
 - invoiceID: int, auto_increment, not NULL, PK.

- customerID: int, FK, not NULL.
- employeeID: int, FK, not NULL.
- invoiceDate: date, not NULL.
- Relationships:
 - 1:M between Invoices and InvoiceSunglasses.
 - 1:M between Employees and Invoices.
 - 1:M between Customers and Invoices.
- Additional Details:

The Invoices table participates in three relationships. Each Invoice is linked to exactly one Customer and exactly one Employee. In addition, Invoice participates in a one-to-many relationship with InvoiceSunglasses.

Sunglasses

- Purpose:
 - Stores the catalog of all sunglasses items for sale.
- Attributes:
 - itemID: int, auto_increment, not NULL, PK.
 - itemName: varchar(100), not NULL.
 - retailPrice: decimal(5,2), not NULL.
 - stockQuantity: int, not NULL, default 0.

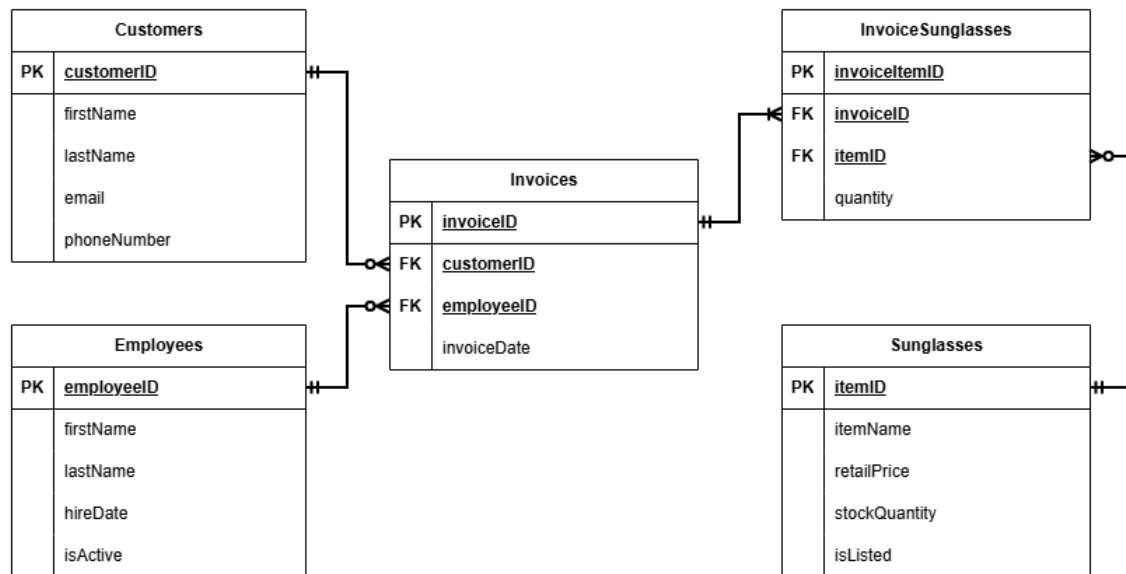
- isListed: tinyint (boolean), not NULL, default 1.
- Constraints:
 - stockQuantity >= 0.
 - retailPrice >= 0.
- Relationships:
 - 1:M between Sunglasses and InvoiceSunglasses.
 - M:N between Sunglasses and Invoices via InvoiceSunglasses.
- Additional Details:
 - Rainbow Sunglasses only sells sunglasses, so in essence the Sunglasses table is our de facto products entity. The itemID attribute is a unique ID for database functions. Sunglasses and Invoices have a many-to-many relationship, implemented through the InvoiceSunglasses intersection table, which creates two one-to-many relationships.
 - All sunglasses have the attribute isListed set to 1 by default, meaning the item exists in the inventory. Upon a soft deletion, the attribute will be set to 0 instead of permanently removing the item.

InvoiceSunglasses

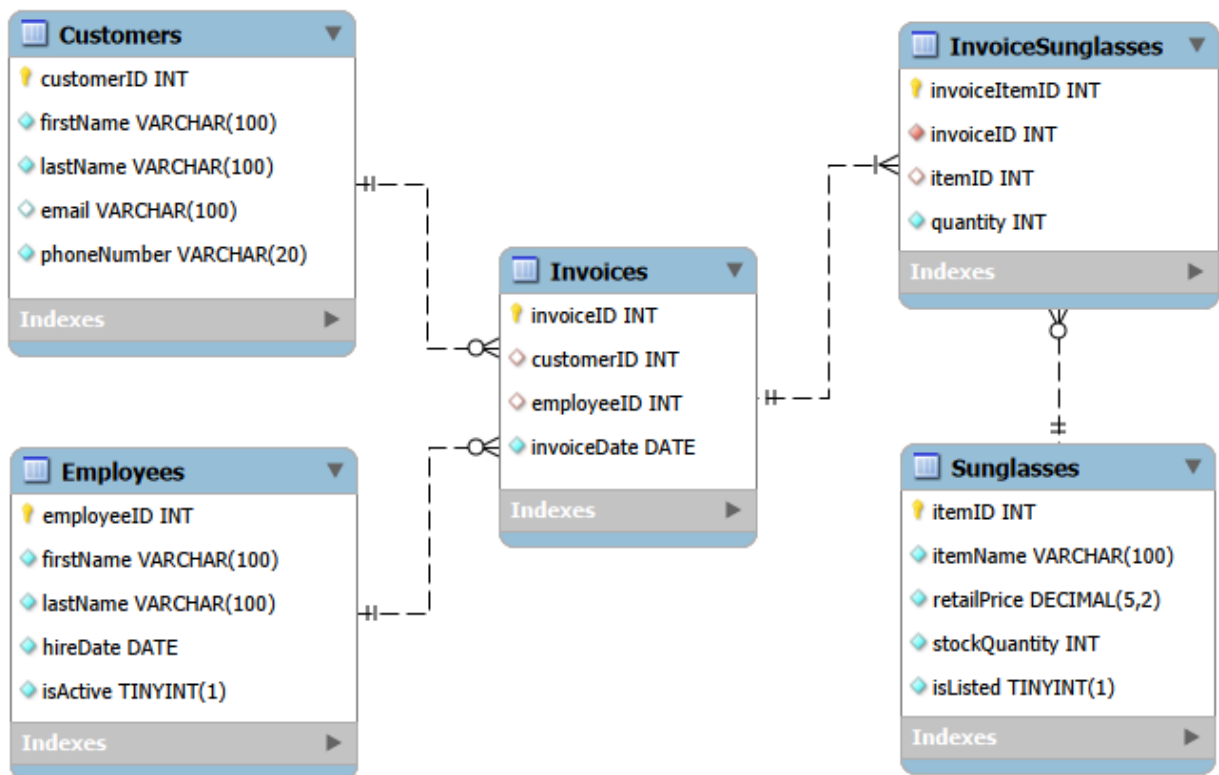
- Purpose:
 - An intersection table to handle the many-to-many relationship.
- Attributes:
 - invoiceItemID: int, auto_increment, not NULL, PK.

- invoiceID: int, FK, not NULL.
- itemID: int, FK, not NULL.
- quantity: int, not NULL.
- Constraints:
 - Composite unique pair (invoiceID, sunglassesID).
 - Quantity > 0.
- Relationships:
 - 1:M between Invoice and InvoiceSunglasses.
 - 1:M between Sunglasses and InvoiceSunglasses.
- Additional Details:
 - This intersection table links to Invoices and Sunglasses via one-to-many relationships (1:M). The quantity attribute is included to record the amount of sunglasses items on an invoice. invoiceSunglassesID acts as a surrogate primary key for each item in this table.

Entity-Relationship Diagram



Schema Diagram



Sample Data

Customers

customerID	firstName	lastName	email	phoneNumber
1	Ava	Nguyen	ava@example.com	503-555-0101
2	Leo	Martinez	NULL	431-645-0102
3	Kevin	Lin	kevin@example.com	713-899-0103
4	Maya	Chen	maya@example.com	385-786-0104

Employees

employeeID	firstName	lastName	hireDate	isActive
1	Steve	Adams	2022-03-23	0
2	Jonny	Patel	2023-01-15	1
3	Jordan	Guy	2024-06-01	1
4	Sasha	Reed	2025-11-20	1

Invoices

invoiceID	customerID	employeeID	invoiceDate
1	2	2	2026-02-01
2	3	3	2026-02-02
3	4	4	2026-02-03
4	2	2	2026-02-04

Sunglasses

itemID	itemName	retailPrice	stockQuantity	isListed
1	Oakley	179.99	0	0
2	Rayban	129.99	12	1
3	Aviator	159.99	8	1
4	Wayfarer	135.99	6	1

5	Retro Square	100.99	7	1
6	Titanium Elite	299.99	10	1
7	ClownGlasses	80.00	5	1

InvoiceSunglasses

invoiceSunglassesID	invoiceID	itemID	quantity
1	1	2	3
2	1	3	5
3	2	2	7
4	3	6	1
5	4	5	1
6	4	7	2

UI Screenshots

READ/BROWSE/DISPLAY **Homepage**



Rainbow Sunglasses

Welcome to the Sunglasses Retail Management System!

Use the navigation bar to manage your store.



Developed by Kevin Lin & Daniel Chambers

READ/BROWSE/DISPLAY **Customers**

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Customers

Customer ID	First Name	Last Name	Email	Phone Number
1	Ava	Nguyen	ava@example.com	503-555-0101
2	Leo	Martinez	N/A	431-645-0102
3	Kevin	Lin	kevin@example.com	713-899-0103
4	Maya	Chen	maya@example.com	385-786-0104

Add Customer

First Name: Last Name: Email: Phone Number:

[Create Customer](#)

Update Customer

Existing Customer: First Name: Last Name: Email: Phone Number:

[Update Customer](#)

READ/BROWSE/DISPLAY **Employees**

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Employees

Employee ID	First Name	Last Name	Hire Date	Active
1	Steve	Adams	2022-03-23	No
2	Jonny	Patel	2023-01-15	Yes
3	Jordan	Guy	2024-06-01	Yes
4	Sasha	Reed	2025-11-20	Yes

Add Employee

First Name: Last Name: Hire Date: Active:

[Create Employee](#)

Update Employee

Existing Employee: First Name: Last Name: Hire Date: Active: [Update Employee](#)

READ/BROWSE/DISPLAY Invoices

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Invoices

Invoice ID	Customer	Employee	Invoice Date	
1	Ava Nguyen	Jonny Patel	2026-02-01	Delete
2	Leo Martinez	Jordan Guy	2026-02-02	Delete
3	Kevin Lin	Sasha Reed	2026-02-03	Delete
4	Maya Chen	Jonny Patel	2026-03-04	Delete

Add Invoice

Customer: Employee: Invoice Date:

Update Invoice

Existing Invoice: Customer: Employee: Date:

READ/BROWSE/DISPLAY Invoice Sunglasses

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[Reset Database](#)

Invoice Sunglasses

Invoice ID	Customer	Sunglasses	Purchase Quantity	
1	Ava Nguyen	Aviator	5	Delete
1	Ava Nguyen	Rayban	3	Delete
2	Leo Martinez	Rayban	7	Delete
3	Kevin Lin	Titanium Elite	1	Delete
4	Maya Chen	Retro Square	1	Delete
4	Maya Chen	ClownGlasses	2	Delete

Add Sunglasses to Invoice

Invoice ID:

Sunglasses:

Quantity:

[Add to Invoice](#)

Update Invoice Sunglasses

Existing Entry:

Sunglasses:

Quantity:

[Update Entry](#)

READ/BROWSE/DISPLAY **Sunglasses**

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Sunglasses

Item ID	Item Name	Retail Price	Stock Quantity	For Sale
2	Rayban	129.99	12	Yes
3	Aviator	159.99	8	Yes
4	Wayfarer	135.99	6	Yes
5	Retro Square	100.99	7	Yes
6	Titanium Elite	299.99	10	Yes
7	ClownGlasses	80.00	5	Yes
1	Oakley	179.99	0	NO

Add Sunglasses

Item Name: Retail Price: Stock Quantity: Listed:

[Create Sunglasses](#)

Update Sunglasses

Existing Item: Item Name: Retail Price: Stock Quantity:

Listed:

[Update Sunglasses](#)

CREATE/INSERT/ADD New **Invoice Sunglasses (M:N)**

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Invoice Sunglasses

Invoice ID	Customer	Sunglasses	Purchase Quantity	
1	Ava Nguyen	Aviator	5	Delete
1	Ava Nguyen	Rayban	3	Delete
2	Leo Martinez	Rayban	7	Delete
3	Kevin Lin	Titanium Elite	1	Delete
4	Maya Chen	Retro Square	1	Delete
4	Maya Chen	Titanium Elite	3	Delete
4	Maya Chen	ClownGlasses	2	Delete



Add Sunglasses to Invoice

Invoice ID: Sunglasses: Quantity:
[Add to Invoice](#)

Update Invoice Sunglasses

Existing Entry: Sunglasses: Quantity:
[Update Entry](#)

EDIT/UPDATE Invoice Sunglasses (M:N)

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Invoice Sunglasses

Invoice ID	Customer	Sunglasses	Purchase Quantity	
1	Ava Nguyen	Aviator	5	Delete
1	Ava Nguyen	Rayban	3	Delete
2	Leo Martinez	Rayban	8	Delete
3	Kevin Lin	Titanium Elite	1	Delete
4	Maya Chen	Retro Square	1	Delete
4	Maya Chen	Titanium Elite	3	Delete
4	Maya Chen	ClownGlasses	2	Delete

Add Sunglasses to Invoice

Invoice ID: Sunglasses: Quantity:

[Add to Invoice](#)

Update Invoice Sunglasses

Existing Entry: Sunglasses: Quantity:

[Update Entry](#)

DELETE Invoice Sunglasses (M:N)

Gmail

Prot

CS Basics

classwork.engr.oregonstate.edu:9124 says

Remove this item from the invoice?

CancelOK

Home | Customers

Reset Database

Invoice Sunglasses

Invoice ID	Customer	Sunglasses	Purchase Quantity	
1	Ava Nguyen	Aviator	5	Delete
1	Ava Nguyen	Rayban	3	Delete
2	Leo Martinez	Rayban	7	Delete
3	Kevin Lin	Titanium Elite	1	Delete
4	Maya Chen	Retro Square	1	Delete
4	Maya Chen	ClownGlasses	2	Delete

Add Sunglasses to Invoice

Invoice ID:

Select Invoice

Sunglasses:

Select Sunglasses

Quantity:

Add to Invoice

Update Invoice Sunglasses

Existing Entry:

Choose an Entry

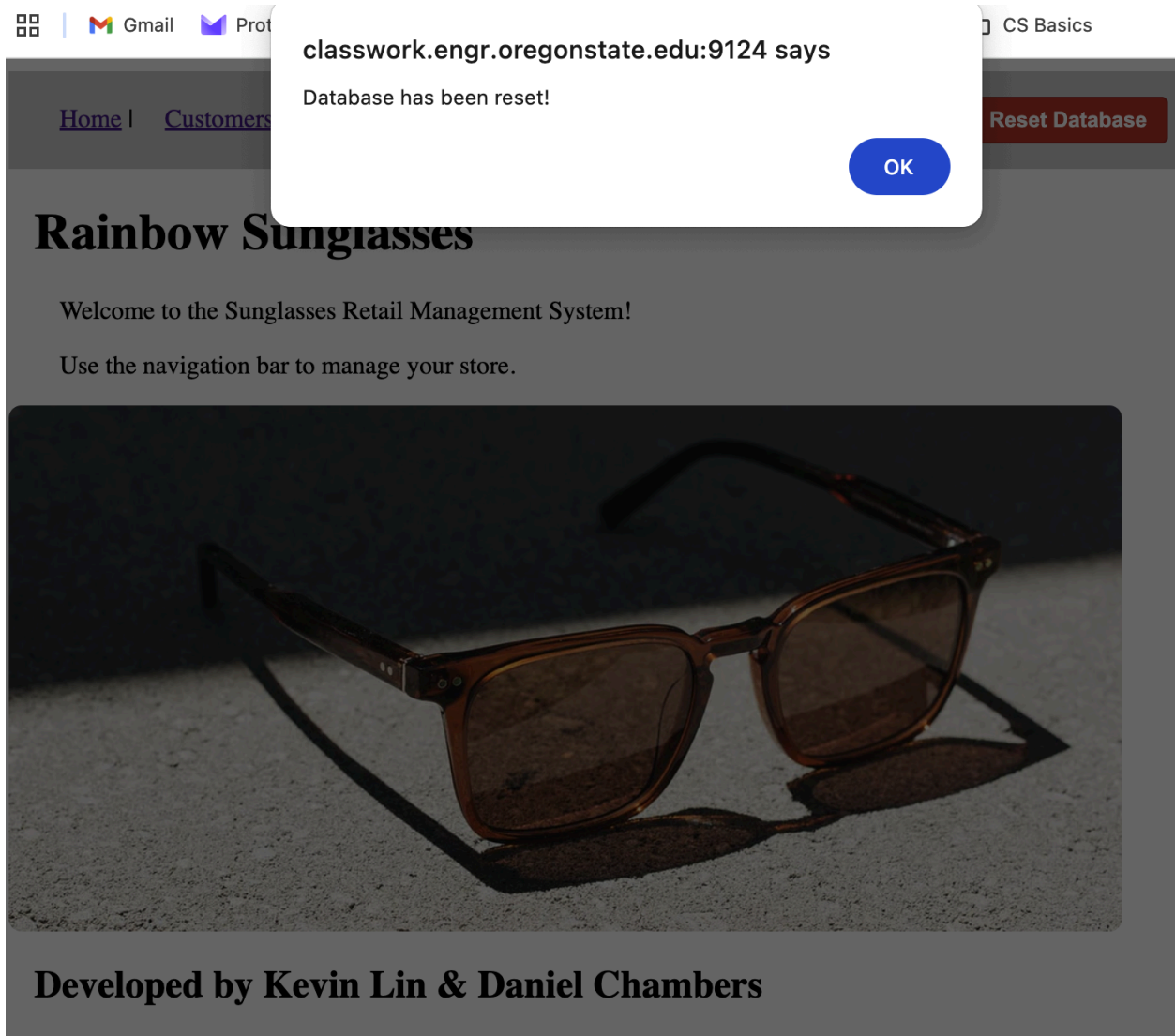
Sunglasses:

Select Sunglasses

Quantity:

Update Entry

RESET DATABASE



Citations / Originality:

This project was developed for CS 340 using course materials, assignment requirements, and instructor-provided examples as references where applicable. AI tools (Copilot) were used in a limited support role for brainstorming, wording refinement, debugging assistance, and documentation support.